

IaC Provisioning for Finance Systems

Report

Course Name: Devops Foundation

Institution Name: Medicaps University – Datagami Skill Based Course

Student Name(s) & Enrolment Number(s):

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Project Number: DO-03

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Academic Year: 2025-2026

Problem Statement & Objectives

1. Problem Statement
2. Project Objectives
3. Scope of the Project

Proposed Solution

1. Key features *(Just mention key features here no need to go into details)*
2. Overall Architecture / Workflow
3. Tools & Technologies Used *(If applicable)*

Results & Output

Add the below details here:

1. Screenshots / outputs
2. Reports / dashboards / models
3. Key outcomes

Conclusion

Mention a brief conclusion about your project summing up everything you have worked on and the key learning.

Future Scope & Enhancements

Problem Statement & Objectives:-

Problem Statement:-

In finance systems, handling a large number of support emails manually is time-consuming and error-prone. There is no automated system to classify, prioritize, and respond to customer queries efficiently. This leads to delayed responses, poor customer experience, and increased workload on support teams.

Project Objectives:-

- To automate email classification using AI
- To reduce manual effort in handling support tickets
- To build a scalable cloud-based system using DevOps
- To implement Infrastructure as Code (IaC)
- To provide a real-time dashboard for monitoring

Scope of the Project:-

- Email fetching and processing
- AI-based classification (Billing, Technical, etc.)
- Data storage and dashboard visualization
- Cloud deployment using AWS EC2
- CI/CD automation using GitHub Actions

Proposed Solution:-

The system provides an automated platform that fetches emails, processes them using AI, and displays categorized results on a dashboard. It uses Docker containers for consistency, Terraform for infrastructure automation, and GitHub Actions for continuous deployment.

Key Features:-

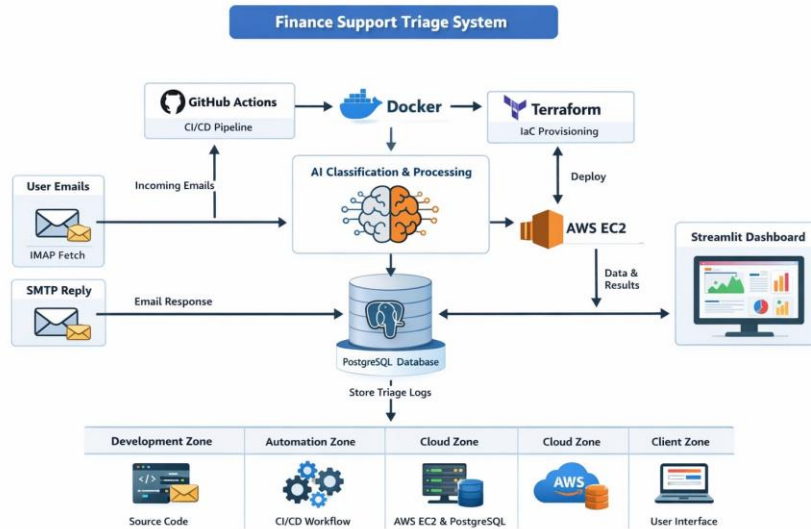
- Automated email fetching (IMAP)
- AI-based classification system
- Real-time dashboard (Streamlit)
- Secure cloud deployment
- CI/CD pipeline automation
- Scalable and fault-tolerant system

Overall Architecture / Workflow:-

Workflow:

1. User sends email
2. System fetches email using IMAP
3. Text is cleaned and processed
4. AI model classifies the email
5. Data is stored in PostgreSQL
6. Dashboard displays results
7. Reply is sent via SMTP

Architecture Layers:

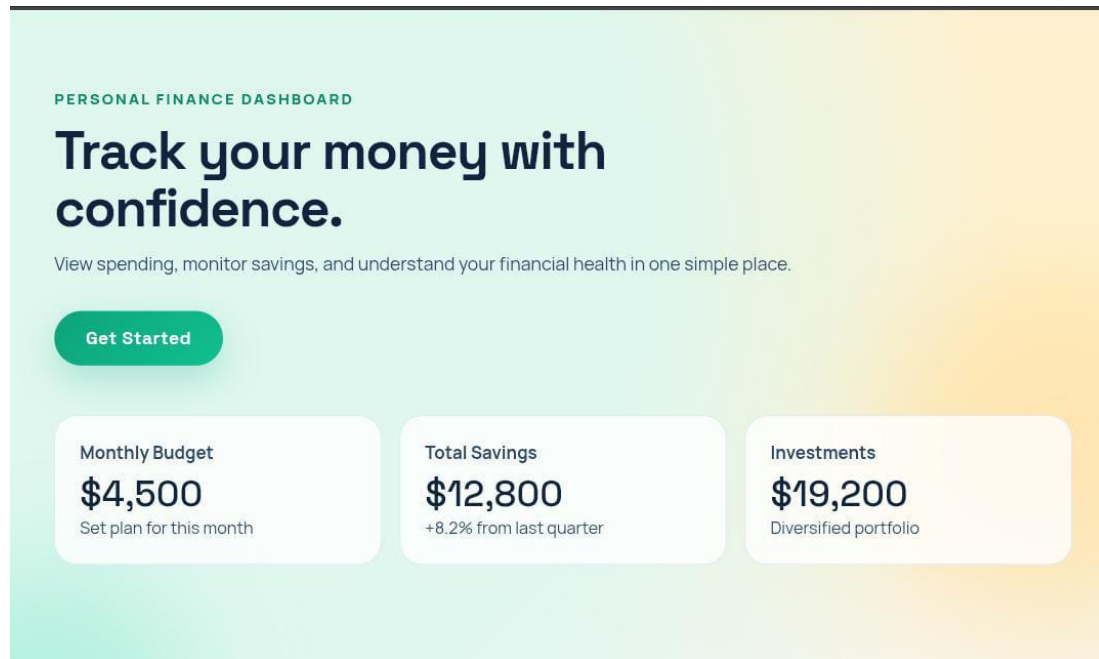


Tools & Technologies Used:-

Frontend:	Streamlit
Backend:	FastAPI, Python
Database:	PostgreSQL (Aiven)
Cloud:	AWS EC2
DevOps:	GitHub Actions
IaC:	Terraform
Containerization:	Docker
Libraries:	SQLAlchemy, imaplib, smtplib

Results & Output:-

Screenshots / Outputs



Key Outcomes:-

- Reduced manual effort by automation
- Faster response time (<2 seconds)
- Improved accuracy in classification
- Scalable and efficient system
- Secure handling of data

Conclusion:-

The Finance Support Triage System successfully automates the process of handling support emails using AI and DevOps practices. It reduces manual workload, improves efficiency, and ensures faster response times. The integration of cloud infrastructure, containerization, and CI/CD pipelines makes the system reliable and scalable.

Future Scope & Enhancements:-

- Integration with real-time ML models
- Multi-language email support
- Advanced analytics dashboard
- Deployment using Kubernetes
- Mobile app integration
- Real-time notifications system